Assignment–5

Design of a 3-Way Traffic Junction Controller Using JK Flip-Flops

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Aim

To design a 3-state traffic junction controller using JK flip-flops with a 60-second clock period and 50% duty cycle.

Traffic Junction Control Principle

In a 3-way traffic junction:

- Only one road must receive the green signal at a time.
- The remaining two roads must remain red.
- The system must operate cyclically and automatically.

Thus, three states are defined:

$$S_0 \to S_1 \to S_2 \to S_0$$

Each state corresponds to one road receiving the green signal while the others remain red.

State Encoding and Memory Requirement

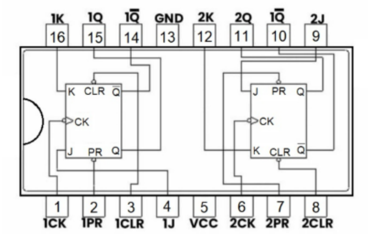
To represent 3 states, binary encoding is required.

$$2^n \geq 3$$

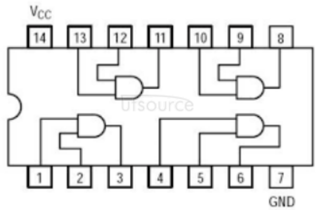
$$2^2 = 4$$

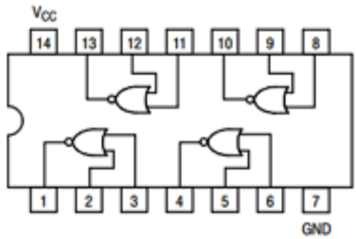
Hence, a minimum of two flip-flops are required to store the states.
The system therefore behaves as a **mod-3 synchronous counter**.

Components Used



1. SN74LS76N
2. SN74LS02N
3. SN74AS08N
4. Green LEDs (3)
5. Red LEDs (3)
6. Resistors (6)
7. Breadboard
8. Arduino Board
9. Connecting Wires





Truth Table

Q_1	Q_0	Q_1^+	Q_0^+	J_0	K_0	J_1	K_1
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	0	0	X	1	0	X
1	1	0	0	X	1	X	1

Derived Input Equations Using K-Maps

Variables: Rows $\rightarrow Q_1$
Columns $\rightarrow Q_0$

J_0

	0	1
0	0	1
1	X	X

$J_0 = Q_0$

K_0

	0	1
0	X	X
1	1	1

$K_0 = I$

J_1

	0	1
0	1	X
1	0	X

$J_1 = \overline{Q_1}$

K_1

	0	1
0	X	1
1	X	1

$K_1 = I$

Clock is connected to Arduino.
VCC = +5V and GND is connected properly.

Output Table

Q_1	Q_0	G_0	R_0	G_1	R_1	G_2	R_2
0	0	1	0	0	1	0	1
0	1	0	1	1	0	0	1
1	0	0	1	0	1	1	0
1	1	X	X	X	X	X	X

Output Logic Expressions

Green Lights

$$G_0 = \overline{Q_1}\,\overline{Q_0}$$
$$G_1 = \overline{Q_1}Q_0$$
$$G_2 = Q_1\overline{Q_0}$$

Red Lights

$$R_0 = \overline{Q_1}Q_0+, Q_1\overline{Q_0} = \overline{G_0}$$
$$R_1 = \overline{Q_0}$$
$$R_2 = \overline{Q_1}$$

Implementation Using AND Gate (74LS08)

- Pin 14 of IC $\rightarrow$ VCC (+5V)
- Pin 7 of IC $\rightarrow$ GND
- Pin 1 and 2 $\rightarrow$ Inputs of AND Gate 1
- Pin 3 $\rightarrow$ Output G_0
- Pin 4 and 5 $\rightarrow$ Inputs of AND Gate 2
- Pin 6 $\rightarrow$ Output G_1
- Pin 9 and 10 $\rightarrow$ Inputs of AND Gate 3
- Pin 8 $\rightarrow$ Output G_2

Implementation Using NOR Gate (74LS02)

- Pin 14 → VCC (+5V)
- Pin 7 → GND
- Output of G_0 connected to NOR gate input
- NOR output gives R_0
- Pin 1 and 2 → Inputs
- Pin 3 → Output

Working of the Circuit

The circuit operates as a synchronous Mod-3 counter using two JK flip-flops.

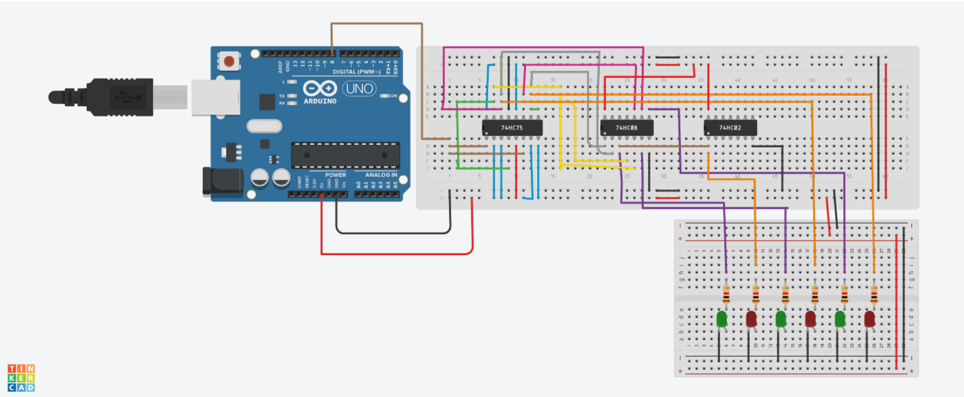
- The Arduino provides a clock pulse of 60 seconds period.
- On every clock pulse, the flip-flops change state according to the derived JK input equations.
- The states cycle as:

$$00 \rightarrow 01 \rightarrow 10 \rightarrow 00$$

- For each state:
  - State 00: Road 0 Green, Road 1 and 2 Red
  - State 01: Road 1 Green, Road 0 and 2 Red
  - State 10: Road 2 Green, Road 0 and 1 Red
- The unused state (11) is automatically forced back to 00, ensuring proper cyclic operation.

Thus, only one green signal is ON at any time, ensuring safe and sequential traffic control.

Circuit Diagram



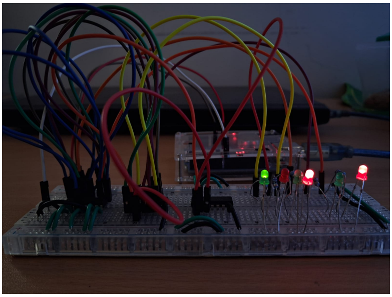
3-Way Traffic Junction Controller Circuit

Arduino Clock Generation Code

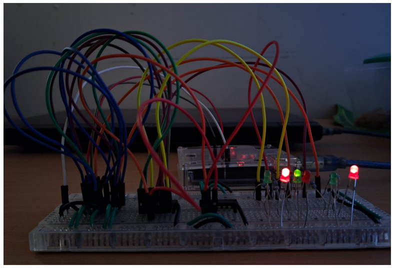
The Arduino is used to generate a periodic clock signal of 60-second period (50% duty cycle) for driving the JK flip-flops.

```
const int clockPin = 8;

void setup() {
  pinMode(clockPin, OUTPUT);
}

void loop() {
  digitalWrite(clockPin, HIGH);
  delay(30000);          * 30 seconds HIGH
  digitalWrite(clockPin, LOW);
  delay(30000);          * 30 seconds LOW
}
```

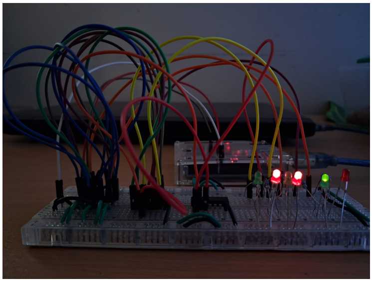
The output from digital pin 8 is connected to the clock inputs of both JK flip-flops.



signal 1



signal 2



signal 3

Issues Faced During Implementation

1. Using Incorrect Datasheet

Initially, an incorrect datasheet was referred for the ICs. The pin configuration and internal structure did not match the actual IC used in the circuit.
Effect: Wrong pin connections were made, and the circuit failed to operate correctly.
Resolution: The correct datasheet corresponding to SN74LS76N, SN74LS02N, and SN74F08N was verified, and all connections were corrected.

2. Incorrect Logic Implementation

During implementation, some logic equations were connected incorrectly.
Effect: The state transition sequence did not follow the required

$$00 \rightarrow 01 \rightarrow 10 \rightarrow 00$$

pattern. At certain states, more than one green LED turned ON.
Resolution: The truth table and K-map simplifications were rechecked, and the logic wiring was corrected accordingly.

3. Internal Breadboard Fault

The breadboard used for implementation had an internal connectivity issue. Some rows were not electrically continuous due to internal strip damage.
Effect: Even though the wiring appeared correct externally, certain nodes were not properly connected, causing inconsistent outputs.
Resolution: The faulty breadboard was replaced, after which the circuit operated stably and correctly.

Conclusion

The 3-way traffic junction controller was successfully designed and implemented using JK flip-flops. The circuit operates as a Mod-3 synchronous counter and generates the required cyclic sequence to control three traffic signals. Practical testing verified correct state transitions and proper operation of the traffic lights.